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Inventor(s)	Brahme; Rohini

HVAC system with improved operation of a single-staged compressor during a peak demand response

Abstract

An HVAC system is configured to regulate a temperature of a space. The HVAC system includes means for compressing a refrigerant used to cool air provided to the space and a controller communicatively coupled to the means for compressing. The controller determines that a demand response time period is starting at a start time. After determining that the demand response time period is starting at the start time, an operation schedule is determined indicating alternating portions of the demand response period during which the means for compressing is to be turned off and turned on. At or after the start time of the demand response time period, the controller begins operating the means for compressing according to the determined operation schedule.

Inventors:	Brahme; Rohini (Irving, TX)
Applicant:	Lennox Industries Inc. (Richardson, TX)
Family ID:	87890104
Assignee:	Lennox Industries Inc. (Richardson, TX)
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Primary Examiner: Nieves; Nelson J

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The application is a continuation of U.S. patent application Ser. No. 17/654,342, filed Mar. 10, 2022, now U.S. Pat. No. 12,078,374 issued Sep. 3, 2024, and entitled “HVAC SYSTEM WITH IMPROVED OPERATION OF A SINGLE-STAGE COMPRESSOR DURING A PEAK DEMAND RESPONSE,” which is incorporated herein by reference.

TECHNICAL FIELD

(1) This disclosure relates generally to heating, ventilation, and air conditioning (HVAC) systems. More particularly, in certain embodiments, this disclosure relates to an HVAC system with improved operation of a single-stage compressor during a peak demand response.

BACKGROUND

(2) Heating, ventilation, and air conditioning (HVAC) systems are used to regulate environmental conditions within an enclosed space. Air is cooled via heat transfer with refrigerant flowing through the HVAC system and returned to the enclosed space as conditioned air.

SUMMARY OF THE DISCLOSURE

(3) In some cases, HVAC systems may be required to operate under restricted operating requirements to reduce power consumption during times of peak electricity demand and/or decreased electricity supply, referred to in this disclosure as peak demand response times or demand response times. For example, a third party such as a utility provider may enforce certain operating restrictions upon HVAC systems during peak demand response times. A peak demand response time may correspond, for example, to a time period associated with high outdoor temperatures or any other time when electrical power consumption is expected (e.g., based on a forecast or projection) to be increased. Generally, the third party (e.g., a utility provider) provides a request, referred to herein as a demand response, which specifies an upper limit on power consumption by an HVAC system during a peak demand response time.

(4) The system of this disclosure solves problems of previous HVAC systems by facilitating improved comfort during peak demand response times by intelligently operating (e.g., turning on and off on a specially determined schedule) a single-stage compressor of an HVAC system more efficiently and effectively than was previously possible. For example, when a demand response is upcoming, a controller of the HVAC system may determine an efficient operation schedule of on and off times for the compressor that improves user comfort during a demand response time while also meeting energy saving requirements. In certain embodiments, the systems and methods described in this disclosure may be integrated into a practical application of an HVAC controller that improves system performance and occupant comfort during peak demand response times by more effectively and efficiently operating the compressor. In certain embodiments, an operation schedule is determined by predicting indoor air temperatures for different possible operation schedules and selecting the operation schedule that most improves occupant comfort while also saving energy (e.g., by meeting comfort and/or energy-saving criteria).

(5) Certain embodiments may include none, some, or all of the above technical advantages. One or more other technical advantages may be readily apparent to one skilled in the art from the figures, descriptions, and claims included herein.

(6) In an embodiment, an HVAC system is configured to regulate a temperature of a space. The HVAC system includes a single-stage compressor configured to compress a refrigerant used to cool air provided to the space and a controller communicatively coupled to the single-stage compressor. The controller determines that a demand response time period is starting at a start time. The demand response time period is a future period of time during which a reduction in energy consumption by the HVAC system is requested. After determining that the demand response time period is starting at the start time, an operation schedule is determined indicating alternating portions of the demand response period during which the single-stage compressor is to be turned off and turned on. At or after the start time of the demand response time period, the controller begins operating the single-stage compressor according to the determined operation schedule.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the present disclosure, reference is now made to the following description, taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a diagram of an example HVAC system configured for improved operation during peak demand response times;

(3) FIG. 2 is a plot illustrating example indoor temperature achieved using the new operation schedule-based approach of this disclosure compared to a temperature achieved using a conventional control strategy;

(4) FIG. 3 is a plot illustrating the indoor temperature and power consumption achieved using

different compressor operating or on times; and

(5) FIG. 4 is a flowchart of an example method of operating the system of FIG. 1.

DETAILED DESCRIPTION

(6) Embodiments of the present disclosure and its advantages are best understood by referring to FIGS. 1 through 4 of the drawings, like numerals being used for like and corresponding parts of the various drawings.

(7) As described above, prior to the present disclosure, there was a lack of tools for improving comfort in a conditioned space in response to a demand response (i.e., a request for decreased HVAC energy consumption). This disclosure recognizes that temperature in a space (e.g., a home, office, or other building) that is serviced by an HVAC system with a single-stage compressor can be maintained at more comfortable levels than was formerly achieved by more efficiently and effectively staging the operations of the compressor. In this way, effective cooling is still provided during the peak demand response time, while still satisfying the energy-saving requirements of the demand response. Turning off a compressor corresponds to stopping or preventing operation of the compressor of the HVAC system, such that the HVAC system does not provide cooling to a corresponding space and such that the energy consumption of the HVAC system is negligible. Likewise, turning on a compressor corresponds to starting or allowing operation of the compressor, such that the HVAC system can provide cooling to the space. For example, when a compressor is turned on, the HVAC system may provide cooling based on a predefined setpoint temperature.

(8) HVAC System

(9) FIG. 1 shows an example HVAC system **100** configured to operate according to a specially determined operation schedule **164** in response to a demand response **140** (abbreviated as “DR” in FIG. 1). A controller **142** of the HVAC system **100** may determine the operation schedule **164**, which indicates times during which the compressor **106** is turned off and on during a demand response time, such that user comfort can be maintained while still achieving the decrease in energy consumption indicated by the demand response **140**. A demand response **140** generally indicates an upper limit on power consumption by the HVAC system **100** during a future period of time (e.g., demand response time **204** of FIG. 2).

(10) The HVAC system **100** conditions air for delivery to a conditioned space (e.g., all or a portion of a room, a house, an office building, a warehouse, or the like). In some embodiments, the HVAC system **100** is a rooftop unit (RTU) that is positioned on the roof of a building and the conditioned air is delivered to the interior of the building. In other embodiments, portion(s) of the system **100** may be located within the building and portion(s) outside the building. The HVAC system **100** may include one or more heating elements, not shown for convenience and clarity. The HVAC system **100** may be configured as shown in FIG. 1 or in any other suitable configuration. For example, the HVAC system **100** may include additional components or may omit one or more components shown in FIG. 1.

(11) The HVAC system **100** includes a working-fluid conduit subsystem **102**, at least one condensing unit **104**, an expansion valve **114**, an evaporator **116**, a blower **128**, and one or more thermostats **136**. The working-fluid conduit subsystem **102** facilitates the movement of a working fluid (e.g., a refrigerant) through a cooling cycle such that the working fluid flows as illustrated by the dashed arrows in FIG. 1. The working fluid may be any acceptable working fluid including, but not limited to hydrofluorocarbons (e.g. R-410A) or any other suitable type of refrigerant.

(12) The condensing unit **104** includes a single-stage compressor **106**, a condenser **108**, and a fan **110**. In some embodiments, the condensing unit **104** is an outdoor unit while other components of system **100** may be located indoors. The compressor **106** is coupled to the working-fluid conduit subsystem **102** and compresses (i.e., increases the pressure of) the working fluid. The compressor **106** is in signal communication with the controller **142** using wired and/or wireless connection. The controller **142** provides commands and/or signals to control operation of the compressor **106** and/or receive signals from the compressor **106** corresponding to a status of the compressor **106**. For

example, the controller **142** may provide signals to turn the compressor **106** on or off based on the operation schedule **164**, which indicates when the single-stage compressor **106** turns on and off during a demand response time.

(13) The condenser **108** is configured to facilitate movement of the working fluid through the working-fluid conduit subsystem **102**. The condenser **108** is generally located downstream of the compressor **106** and is configured to remove heat from the working fluid. The fan **110** is configured to move air **112** across the condenser **108**. For example, the fan **110** may be configured to blow outside air through the condenser **108** to help cool the working fluid flowing therethrough. The fan **110** may be in communication with the controller **142** (e.g., via wired and/or wireless communication) to receive control signals for turning the fan **110** on and off and/or adjusting a speed of the fan **110**. The compressed, cooled working fluid flows from the condenser **108** toward the expansion valve **114**. The fan **110** may be turned on and off along with the compressor **106** based on the operation schedule **164**.

(14) The expansion valve **114** is coupled to the working-fluid conduit subsystem **102** downstream of the condenser **108** and is configured to remove pressure from the working fluid. In this way, the working fluid is delivered to the evaporator **116**. In general, the expansion valve **114** may be a valve such as an expansion valve or a flow control valve (e.g., a thermostatic expansion valve (TXV)) or any other suitable valve for removing pressure from the working fluid while, optionally, providing control of the rate of flow of the working fluid. The expansion valve **114** may be in communication with the controller **142** (e.g., via wired and/or wireless communication) to receive control signals for opening and/or closing associated valves and/or to provide flow measurement signals corresponding to the rate of working fluid flow through the working-fluid conduit subsystem **102**.

(15) The evaporator **116** is generally any heat exchanger configured to provide heat transfer between air flowing through (or across) the evaporator **116** (i.e., airflow **118** contacting an outer surface of one or more coils of the evaporator **116**) and working fluid passing through the interior of the evaporator **116**. The evaporator **116** may include one or more circuits of coils. The evaporator **116** is fluidically connected to the compressor **106**, such that working fluid generally flows from the evaporator **116** to the condensing unit **104** when the HVAC system **100** is operating to provide cooling.

(16) A portion of the HVAC system **100** is configured to move airflow **118** provided by the blower **128** across the evaporator **116** and out of the duct sub-system **122** as conditioned airflow **120**. Return air **124**, which may be air returning from the building, fresh air from outside, or some combination, is pulled into a return duct **126**. A suction side of the blower **128** pulls the return air **124**. The blower **128** discharges airflow **118** into a duct **130** such that airflow **118** crosses the evaporator **116** or heating elements (not shown) to produce conditioned airflow **120**. The blower **128** is any mechanism for providing airflow **118** through the HVAC system **100**. For example, the blower **128** may be a constant-speed or variable-speed circulation blower or fan. Examples of a variable-speed blower include, but are not limited to, belt-drive blowers controlled by inverters, direct-drive blowers with electronic commuted motors (ECM), or any other suitable type of blower.

(17) The HVAC system **100** includes one or more sensors **132**, **134** in signal communication with the controller **142** (e.g., via wired and/or wireless connection). Sensor **132** is positioned and configured to measure an indoor air temperature **154**. Sensor **134** is positioned and configured to measure an occupancy **156** of the space serviced by the HVAC system **100**. For example, an occupancy sensor **134** may be a motion sensor or the like. In some cases, occupancy **156** may be determined using known positions of occupants of the space. For example, geofencing may be used to determine occupancy based on the locations of mobile devices operated by occupants of the space. The HVAC system **100** may include one or more further sensors (not shown for conciseness), such as sensors for measuring air humidity and/or any other properties of a conditioned space (e.g. a room of the conditioned space). Sensors **132**, **134** and/or any other

sensors may be positioned anywhere within the conditioned space, the HVAC system **100**, and/or the surrounding environment.

(18) The thermostat **136** may be located within the conditioned space (e.g. a room or building) serviced by the HVAC system **100**. The controller **142** may be separate from or integrated within the thermostat **136**. The thermostat **136** is configured to allow a user to input a desired temperature or temperature setpoint **138** for the conditioned space. In some embodiments, the thermostat **136** includes a user interface and display for displaying information related to the operation and/or status of the HVAC system **100**. For example, the user interface may display operational, diagnostic, and/or status messages and provide a visual interface that allows at least one of an installer, a user, a support entity, and a service provider to perform actions with respect to the HVAC system **100**. For example, the user interface may provide for display of messages related to the status and/or operation of the HVAC system **100** (e.g., whether the HVAC system **100** is being operated for a demand response **140** according to an operation schedule **164** determined by the controller **142**). The thermostat **136** may further be configured to monitor a historical power consumption of the HVAC system **100**, which may be used to generate the home model **158**, as described further below.

(19) The thermostat **136** (and/or controller **142**) may be in communication with a utility provider or other third party tasked with overseeing and/or regulating energy consumption by the HVAC systems **100**. For example, a utility provider or third party may be a company or organization that distributes energy to homes and businesses. In situations in which energy demand is anticipated to exceed supply, a demand response **140** may be transmitted to HVAC system **100**. As described above, the demand response **140** indicates a prescribed reduction in energy consumption (e.g., a percent reduction in energy consumption from a baseline or average value) or a maximum energy consumption (e.g., a maximum permitted energy consumption per time) during the future period of time during which a decrease in energy consumption is needed.

(20) The controller **142** is communicatively coupled (e.g., via wired and/or wireless connection) to components of the HVAC system **100** and configured to control their operation. The controller **142** generally determines that a demand response **140** has been received and that a time period (e.g., time period **204** of FIG. 2) is upcoming during which a reduction in energy consumption by the HVAC system **100** is requested. The controller **142** then determines an operation schedule **164** indicating alternating portions of the demand response period during which the single-stage compressor **106** is to be turned off and turned on. The operation schedule **164** indicates the distribution of on and off times of single-stage compressor **106** over the time interval of the demand response **140**. For example, the operation schedule **164** may indicate period of times during the demand response time during which the compressor **106** is on or off (see, e.g., alternating on times **212a-d** and off times **210a-d** during time period **204** of FIG. 2). In some cases, the operation schedule **164** may indicate an alternative energy-saving setpoint at which to operate the HVAC system **100** (e.g., comfort setpoint **202** of FIG. 2). For example, if the HVAC system **100** is cooling an occupied space, the setpoint **138** may be “setback” to a higher temperature for cooling mode operation.

(21) In some embodiments, the controller **142** determines the operation schedule **164** using a home model **158**. The home model **158** generally allows predicted indoor temperature(s) **160** and/or predicted occupancy **162** to be determined for the space serviced by HVAC system **100** for different possible operation scenarios **152**. For instance, the home model **158** may allow different operation scenarios **152** to be proactively tested and refined to further improve occupant comfort via determination of predicted indoor temperature(s) **160** and/or predicted occupancy **162**. The operation scenarios **152** are different on and off timings/schedules for the single-stage compressor **106**. A predicted indoor temperature **160** may be determined (e.g., as a temperature over time, as shown for temperature **208** of FIG. 2) for each operation scenario **152**.

(22) The home model **158** may also account for a temperature forecast **150**, the current indoor

temperature **154**, and/or an occupancy **156** of the space serviced by the HVAC system **100**. For example, the home model **158** may be used to determine predicted indoor air temperature **160** as a function of one or more of outdoor air temperature (e.g., using a temperature forecast **150**), compressor on/off status (e.g., from the operation scenario **152**), occupancy (e.g., using a measured occupancy **156** and/or predicted occupancy **162**), and a length of the demand response time period. The operation schedule **164** is the operation scenario **152** with the best performance (e.g., the operation scenario **152** for which the predicted indoor air temperature **160** is less than a threshold comfort value (e.g., temperature difference threshold **306** of FIG. 3). In some cases, the operation schedule **164** is determined as the operation scenario **152** with a predicted indoor air temperature **160** that is less than the threshold comfort value (e.g., temperature difference threshold **306** of FIG. 3) and with an energy consumption that is less than a predefined energy consumption (e.g., energy consumption threshold **308** of FIG. 3).

(23) The home model **158** may be determined, for example, based at least in part on historical power consumption of the HVAC system **100** and historical indoor temperatures achieved by the HVAC system **100**. For example, historical information about power consumption by the HVAC system **100** may be used to generate the home model **158** and subsequently determines predicted indoor temperatures **160** for a given operation scenario **152**. The home model **158** may provide predicted indoor temperature **160** and/or predicted occupancy **162** as a function of run time (e.g., as indicated in operation schedule **164**). One or more rounds of iteration may be used to test and/or adjust the operation scenarios **152** to determine the operation schedule **164** that maintains the predicted indoor temperature(s) **160** in a target range (see, e.g., the maintenance of temperature **208** below the comfort temperature value/setpoint **202** using operation schedule **164** and thresholds **306**, **308** of FIG. 3). Examples of home models **158** and their development are described in U.S. Pat. No. 10,612,804, entitled “Operating an HVAC system to reach target temperature efficiently”; U.S. Pat. No. 10,612,808, entitled “Operating an HVAC system based on predicted indoor air temperature”; U.S. Pat. No. 10,830,474, entitled “Systems and methods of predicting energy usage”; and U.S. Pat. No. 11,067,305, entitled “Method and system for heating auto-setback”, each of which is incorporated herein in its entirety.

(24) Once the operation schedule **164** is determined, controller **142** causes the compressor **106** to operate according to the operation schedule **164**. For example, the controller **142** may send signals at the start time of the demand response causing the compressor **106** to turn on and off according to the operation schedule **164**. During the demand response time **204**, the controller **142** may override operation according to the temperature setpoint **138**, such that the HVAC system **100** is not operated according to the temperature setpoint **138** during at least a portion of the demand response time period. Instead, the controller **142** may give preference to turning on and off the compressor **106** according to the operation schedule **164**.

(25) In some cases, the controller **142**, while operating the compressor **106** according to the operation schedule **164**, may determine that the indoor air temperature **154** becomes greater than a predefined maximum temperature (e.g., the comfort temperature setpoint **202** of FIG. 2) and increase cooling to bring the indoor air temperature **154** below this level. For example, after determining that the indoor air temperature **154** is greater than the predefined maximum temperature, the controller **142** may cause the single-stage compressor **106** to turn on at least until the indoor air temperature **154** becomes less than the predefined maximum temperature. In this way, the demand response **140** may be briefly paused to ensure that the space remains comfortable for occupants.

(26) In some embodiments, the operation schedule **164** may be adjusted based at least in part on occupancy **156** of the space cooled by the HVAC system **100** and or a predicted occupancy **162** determined by the home model **158**. Occupancies **156**, **162** may be “occupied” if one or more people are in the space (or predicted to be in the space) or “unoccupied” if no one is in the space (or no one is predicted to be in the space). An occupancy sensor **134** may be used to determine the

occupancy **156**. Historical values of the occupancy **156** may be used by the home model **158** to determine the predicted occupancy **162**, which may indicate that a space is likely to be occupied during certain portions of the day and unoccupied during other portions of the day. If the space serviced by the HVAC system **100** becomes unoccupied during the demand response time or is predicted to be unoccupied during the demand response time, the operation schedule **164** may be adjusted to cause compressor **106** to shut off at least when the serviced space is unoccupied.

(27) The controller **142** includes a processor **144**, memory **146**, and input/output (I/O) interface **148**. The processor **144** comprises one or more processors operably coupled to the memory **146**. The processor **144** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g. a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs) that communicatively couples to memory **146** and controls the operation of HVAC system **100**. The processor **144** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **144** is communicatively coupled to and in signal communication with the memory **146**. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **144** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **144** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory **146** and executes them by directing the coordinated operations of the ALU, registers, and other components. The processor may include other hardware and software that operates to process information, control the HVAC system **100**, and perform any of the functions described herein (e.g., with respect to FIGS. 1-4). The processor **144** is not limited to a single processing device and may encompass multiple processing devices.

(28) The memory **146** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **146** may be volatile or non-volatile and may comprise ROM, RAM, ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). The memory **146** is operable to store any suitable set of instructions, logic, rules, and/or code for executing the functions described in this disclosure with respect to FIGS. 1-4. The memory **146** may store the temperature forecast **150**, operation scenarios **152**, indoor temperatures **154**, occupancies **156**, the home model **158**, predicted indoor temperatures **160**, predicted occupancies **162**, and operation schedules **164**.

(29) The I/O interface **148** is configured to communicate data and signals with other devices. For example, the I/O interface **148** may be configured to communicate electrical signals with the other components of the HVAC systems **100**. The I/O interface **148** may send signals that cause the operation schedule **164** to be implemented by the compressor **106**. The I/O interface **148** may use any suitable type communication protocol. The I/O interface **148** may comprise ports and/or terminals for establishing signal communications between the controller **142** and other devices. The I/O interface **148** may be configured to enable wired and/or wireless communications.

(30) Connections between various components of the HVAC system **100** and between components of system **100** may be wired or wireless. For example, conventional cable and contacts may be used to couple the thermostat **136** to the controller **142** and various components of the HVAC system **100**, including, the compressor **106**, the expansion valve **114**, the blower **128**, and/or sensor(s) **132**, **134**. In some embodiments, a wireless connection is employed to provide at least some of the connections between components of the HVAC system **100**. In some embodiments, a data bus couples various components of the HVAC system **100** together such that data is communicated there between. In a typical embodiment, the data bus may include, for example, any combination of

hardware, software embedded in a computer readable medium, or encoded logic incorporated in hardware or otherwise stored (e.g., firmware) to couple components of HVAC system **100** to each other.

(31) As an example and not by way of limitation, the data bus may include an Accelerated Graphics Port (AGP) or other graphics bus, a Controller Area Network (CAN) bus, a front-side bus (FSB), a HYPERTRANSPORT (HT) interconnect, an INFINIBAND interconnect, a low-pin-count (LPC) bus, a memory bus, a Micro Channel Architecture (MCA) bus, a Peripheral Component Interconnect (PCI) bus, a PCI-Express (PCI-X) bus, a serial advanced technology attachment (SATA) bus, a Video Electronics Standards Association local (VLB) bus, or any other suitable bus or a combination of two or more of these. In various embodiments, the data bus may include any number, type, or configuration of data buses, where appropriate. In certain embodiments, one or more data buses (which may each include an address bus and a data bus) may couple the controller **142** to other components of the HVAC system **100**.

(32) In an example operation of the system **100**, the compressor **106** of the HVAC system **100** is initially turned on and allowed to facilitate cooling of the space based on the temperature setpoint **138**. A demand response **140** is then received indicating that a decrease in energy consumption is needed during an upcoming period of time. An operation schedule **164** is then determined for turning the compressor **106** on and off during the demand response time of the demand response **140**, and the compressor **106** is operated according to the operation schedule **164**.

(33) FIG. 2 shows a plot **200** illustrating an example temperature **208** that may be achieved during a demand response **140** using the improved operation schedule **164** compared to a temperature **206** achieved using a conventional approach. The example operation schedule **164** designates alternating off times **210a-d** and on times **212a-d** for the compressor **106** during the time period **204** of the demand response **140**.

(34) In the conventional approach, an increased comfort setpoint **202** is used during time period **204**, such that the temperature **206** increases from the initial setpoint **138** until it exceeds the comfort setpoint **202** and the compressor **106** needs to turn on for a relatively long period of time to bring the temperature back below the comfort setpoint **202**. In the example of FIG. 2, temperature **206** never reaches the comfortable range below comfort setpoint **202** during the demand response time **204**, such that the space is uncomfortably warm during the majority of the demand response time **204**.

(35) In the new approach of this disclosure using the operation schedule **164**, the temperature **208** increases a relatively small amount during each off time **210a-d** and decreases during each on time **212a-d** when cooling is provided to the space. In this way, temperature **208** is maintained in a more comfortable range while still satisfying energy consumption requirements of the demand response **140**.

(36) FIG. 3 is a plot **300** illustrating how the temperature difference **302** between the indoor temperature **154** and the original setpoint **138** varies with the on time of the compressor **106** during each fifteen-minute period of the demand response time **204** of FIG. 2. As the on time increases (from zero minutes per fifteen-minute interval to eleven minutes per fifteen-minute interval), the temperature difference **302** decreases. A decreased temperature difference **302** corresponds to improved comfort in the space serviced by the HVAC system **100**. At on times of four minutes and greater, the temperature difference **302** is less than a comfort threshold value **306** corresponding to adequate comfort in the space.

(37) Plot **300** of FIG. 3 also shows the percentage power usage **304** of the HVAC system **100** at different on times. A percentage power usage **304** of 100% corresponds to the scenario in which the compressor **106** is turned on for the entire fifteen-minute interval. The percentage power usage **304** increases with the length of the on time. At on times of seven minutes and less, the percentage power usage **304** is less than a threshold value **308** corresponding to a maximum power usage allowed for the demand response **140**. Accordingly, on times of four minutes and seven minutes

both satisfy the energy-saving requirements of the demand response **140** (i.e., by keeping percentage power usage **304** below threshold **308**) and maintain comfort in the space (e.g., by maintaining the temperature difference **302** below threshold **306**). As such, an operation schedule **164** may be determined based on the plot **300** in which the on times **212a-d** are between about four minutes and seven minutes of each fifteen-minute interval of the demand response time **204**.

(38) Example Method of Operation

(39) FIG. **4** is a flowchart of an example method **400** of operating the system of FIG. **1**. Steps of method **400** may be implemented using the processor **144**, memory **146**, and I/O interface **148** of the controller **142**. In some cases, one or more steps may be performed by other components of the system **100** (e.g., by the thermostat **136**). Method **400** may begin at step **402** where it is determined whether the start time of a demand response **140** is upcoming. If a demand response time (e.g., time **204** of FIG. **2**) is upcoming, the controller **142** proceeds to step **404**. Otherwise, the controller **142** waits until a demand response **140** is received indicating an upcoming demand response time.

(40) At step **404**, the controller **142** receives temperature forecast **150**. The temperature forecast **150** may include predicted future outdoor air temperatures for the location in which the HVAC system **100** is operated. The temperature forecast **150** may be obtained from a weather forecast for the location of the HVAC system **100**.

(41) At step **406**, the controller **142** determines predicted indoor air temperature and/or predicted occupancy for one or more operation scenarios **152**. For example, the home model **158** of FIG. **1** may be used to determine values of the predicted indoor air temperature **160** over time for the predicted outdoor air temperatures indicated by the temperature forecast **150** when the compressor **106** is turned on and off according to the operation scenarios **152**.

(42) At step **408**, the controller **142** determines the operation schedule **164** based on information from step **406**. For example, the determined operation schedule **164** may be the operation scenario **152** for which the predicted indoor air temperature **160** is less than a comfort threshold value and the energy consumption is less than a predefined energy consumption threshold. For instance, referring to the example of FIG. **3**, the determined operation schedule **164** may have an on time that satisfies both temperature difference threshold **306** and energy consumption threshold **308**.

(43) At step **410**, the controller **142** operates the compressor **106** according to the operation schedule **164** from step **408**. The controller **142** causes the compressor **106** to turn off and on during the different periods of time determined at step **408**.

(44) Modifications, additions, or omissions may be made to method **400** depicted in FIG. **4**. Method **400** may include more, fewer, or other steps. For example, steps may be performed in parallel or in any suitable order. While at times discussed as the controller **142** performing the steps, any suitable components (e.g., thermostat **136**) of the system **100** may perform one or more steps of the method **400**.

(45) While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated in another system or certain features may be omitted, or not implemented.

(46) In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

(47) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112 (f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A heating, ventilation, and air conditioning (HVAC) system configured to regulate a temperature of a space, the HVAC system comprising: means for compressing a refrigerant used to cool air provided to the space; a controller communicatively coupled to the means for compressing the refrigerant, the controller configured to: determine that a demand response time period is starting at a start time, wherein the demand response time period is a future period of time during which a reduction in energy consumption by the HVAC system is requested; after determining that the demand response time period is starting at the start time, determine an operation schedule indicating alternating portions of the demand response time period during which the means for compressing the refrigerant is to be turned off and turned on; and at the start time of the demand response time period, begin operating the means for compressing the refrigerant according to the determined operation schedule.
2. The HVAC system of claim 1, wherein the controller is further configured to determine the operation schedule by: determining, using a predetermined home model, a predicted indoor air temperature for each of a set of predefined operation scenarios, wherein the home model indicates predicted indoor air temperature as a function of one or more of outdoor air temperature, compressor on/off status, and a length of the demand response time period; and determining the operation schedule as a selected operation scenario from the set of predefined operation scenarios with the predicted indoor air temperature that is less than a threshold comfort value.
3. The HVAC system of claim 2, wherein the controller is further configured to determine the operation schedule as the selected operation scenario with the predicted indoor air temperature that is less than the threshold comfort value and with an energy consumption that is less than a predefined energy consumption threshold value.
4. The HVAC system of claim 2, wherein the controller is further configured to determine the predicted indoor air temperature for each of the predefined operation scenarios using information from an outdoor temperature forecast.
5. The HVAC system of claim 2, wherein the controller is further configured to: determine a predicted occupancy of the space during the demand response time period; and determine that the means for compressing the refrigerant is turned off at least during portions of the demand response time period that the space is predicted to be unoccupied based on the predicted occupancy.
6. The HVAC system of claim 1, wherein the controller is further configured to override operation according to a temperature setpoint during at least a portion of the demand response time period.
7. The HVAC system of claim 1, wherein the controller is further configured to, while the means for compressing the refrigerant is operating according to the operation schedule: determine that an indoor air temperature is greater than a predefined maximum temperature; and after determining that the indoor air temperature is greater than the predefined maximum temperature, cause the means for compressing the refrigerant to turn on at least until the indoor air temperature is less than the predefined maximum temperature.
8. A method of operating a heating, ventilation, and air conditioning (HVAC) system configured to regulate a temperature of a space, the method comprising: determining that a demand response time period is starting at a start time, wherein the demand response time period is a future period of time during which a reduction in energy consumption by the HVAC system is requested; after determining that the demand response time period is starting at the start time, determining an operation schedule indicating alternating portions of the demand response time period during which

a means for compressing a refrigerant of the HVAC system is to be turned off and turned on; and at the start time of the demand response time period, operating the means for compressing a refrigerant according to the determined operation schedule.

9. The method of claim 8, further comprising determining the operation schedule by: determining, using a predetermined home model, a predicted indoor air temperature for each of a set of predefined operation scenarios, wherein the home model indicates predicted indoor air temperature as a function of one or more of outdoor air temperature, compressor on/off status, and a length of the demand response time period; and determining the operation schedule as a selected operation scenario from the set of predefined operation scenarios with the predicted indoor air temperature that is less than a threshold comfort value.

10. The method of claim 9, further comprising determining the operation schedule as the selected operation scenario with the predicted indoor air temperature that is less than the threshold comfort value and with an energy consumption that is less than a predefined energy consumption threshold value.

11. The method of claim 9, further comprising determining the predicted indoor air temperature for each of the predefined operation scenarios using information from an outdoor temperature forecast.

12. The method of claim 9, further comprising: determining a predicted occupancy of the space during the demand response time period; and determining that the means for compressing the refrigerant is turned off at least during portions of the demand response time period that the space is predicted to be unoccupied based on the predicted occupancy.

13. The method of claim 8, further comprising overriding operation according to a temperature setpoint during at least a portion of the demand response time period.

14. The method of claim 8, further comprising, while the means for compressing the refrigerant is operating according to the operation schedule: determining that an indoor air temperature is greater than a predefined maximum temperature; and after determining that the indoor air temperature is greater than the predefined maximum temperature, causing the means for compressing the refrigerant to turn on at least until the indoor air temperature is less than the predefined maximum temperature.

15. A controller of a heating, ventilation, and air conditioning (HVAC) system, the controller comprising: an interface communicatively coupled to a means for compressing a refrigerant configured to compress a refrigerant used to cool air provided to a space; and a processor communicatively coupled to the interface, the processor configured to: determine that a demand response time period is starting at a start time, wherein the demand response time period is a future period of time during which a reduction in energy consumption by the HVAC system is requested; after determining that the demand response time period is starting at the start time, determine an operation schedule indicating alternating portions of the demand response time period during which the means for compressing the refrigerant is to be turned off and turned on; and at the start time of the demand response time period, begin operating the means for compressing the refrigerant according to the determined operation schedule.

16. The controller of claim 15, wherein the processor is further configured to determine the operation schedule by: determining, using a predetermined home model, a predicted indoor air temperature for each of a set of predefined operation scenarios, wherein the home model indicates predicted indoor air temperature as a function of one or more of outdoor air temperature, compressor on/off status, and a length of the demand response time period; and determining the operation schedule as a selected operation scenario from the set of predefined operation scenarios with the predicted indoor air temperature that is less than a threshold comfort value.

17. The controller of claim 16, wherein the processor is further configured to determine the selected operation schedule as the operation scenario with the predicted indoor air temperature that is less than the threshold comfort value and with an energy consumption that is less than a predefined energy consumption threshold value.

18. The controller of claim 16, wherein the processor is further configured to determine the predicted indoor air temperature for each of the predefined operation scenarios using information from an outdoor temperature forecast.

19. The controller of claim 16, wherein the processor is further configured to: determine a predicted occupancy of the space during the demand response time period; and determine that the means for compressing the refrigerant is turned off at least during portions of the demand response time period that the space is predicted to be unoccupied based on the predicted occupancy.

20. The controller of claim 15, wherein the processor is further configured to, while the means for compressing the refrigerant is operating according to the operation schedule: determine that an indoor air temperature is greater than a predefined maximum temperature; and after determining that the indoor air temperature is greater than the predefined maximum temperature, cause the means for compressing the refrigerant to turn on at least until the indoor air temperature is less than the predefined maximum temperature.
